Worksheet A: Research Design

Name: _____ Date: _____

Task 1: Case study: analysing a design

in class, pairs, 15 min.

The following study is fictitious and was written for this exercise; all numbers are invented.

"Learning app improves maths grades by 0.7"

In the winter semester, a university offers a new practice app for the lecture Mathematics 1. The app was developed by the lecturer himself. Use is voluntary; 64 of 212 students sign up. To introduce the app, the lecturer offers the registered students a weekly tutorial in which they practise with the app. At the end of the semester, all 212 students take the same exam, which the lecturer marks himself; name and student number are on each script. The app group achieves an average grade of 2.3, the other 148 students 3.0 (German scale: 1 = best, 5 = fail). The university's press release says: "The app improves the grade by 0.7. From next semester it will be compulsory for all first-year students."

1. Name the IV and DV (independent and dependent variable). How was the DV measured?
2. What type of design is this? Write the design in R/X/O notation (one line per group; R = random assignment, X = treatment, O = observation).
3. Is there a control group? Apart from using the app, in which respects does it differ from the app group?
4. Name at least three confounders or threats to internal validity and match them to technical terms (e.g. selection, history, maturation, testing, instrumentation, attrition, researcher expectations).
5. Is the measurement objective? What would a blind design look like?
6. Sketch an improved design that could test the claim in the press release.

Task 2: Design sketch for your hypothesis

start in class, due by Unit 8, graded

Fill in the template and justify your choices on at most one further page.

H_1 (from Unit 5)	
IV and its levels	
DV and measuring instrument	
Type of design (experimental / quasi / non-experimental; cross-sectional or longitudinal)	
Design in notation (R / X / O)	
Sample: who, how many, how recruited?	
Analysis plan (from Unit 6): test or Bayes factor, smallest effect of interest, n from the power table	
Assignment to conditions	
Control condition (active?)	
Main confounders (dragons on your map from Unit 6) and how I control them	
Measurement quality: objective? reliable? valid? blind? parallel measurement?	
Remaining threat that I accept	